

Tutorial Sheet

Topic: Trees

Question 1

- A. Draw three different binary trees of 7 nodes each that satisfy the following conditions: A Full Binary Tree, A Complete Binary Tree

For each of the above trees, mention: The height of the tree, The number of leaf nodes. Whether it can also be a Perfect Binary Tree (justify your answer briefly)

- B. Construct a binary tree using the following traversal sequences: Inorder: D B E A F C, Preorder: A B D E C F. Write the Postorder traversal of the constructed tree.

Question 2

- A. Create a Binary Search Tree (BST) by inserting the keys in this order: 45, 10, 90, 7, 12, 50, 80. Delete the node with value 10 from the BST and draw the resulting tree.
- B. Insert the following elements into an initially empty BST: 50, 30, 70, 20, 40, 60, 80.

Question 3

- A. Compare AVL Tree, Red-Black Tree, and **B**-Tree based on balancing mechanism and applications. Mention two real-world applications of binary trees and explain briefly how trees are useful in each.
- B. Insert the following sequence of keys into an AVL Tree: 30, 20, 40, 10, 25, 35, 50 . Show the rotations (if any) performed to maintain balance.
- C. Starting with an empty tree, insert the following keys into a Red-Black Tree in the given order: 10, 20, 30, 15, 25. Show the tree after each insertion and indicate the rotations and colour changes needed to maintain Red-Black properties.